

Sequential Decision Making: Theory and Practice

Lecture 1: Logistics and Stochastic Bandits

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Yale Statistics & Data Science



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Outline

Course introduction

Logistics

Stochastic bandits

Supervised learning

Given training data, make prediction on unseen data:

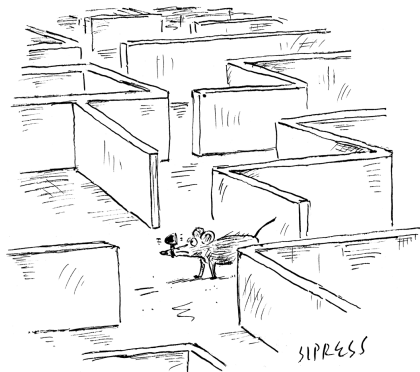


Primarily deal with **pattern recognition**

Reinforcement learning

In RL, an agent learns by interacting with an environment.

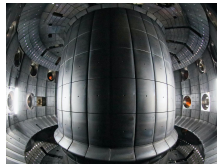
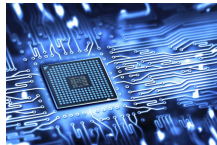
- no training data
- maximize total rewards
- trial-and-error
- sequential and online



"Recalculating ... recalculating ..."

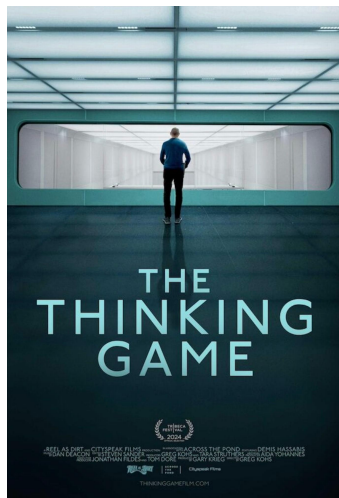
Deal with **decision making**, sometimes with constraints

Recent successes in reinforcement learning (RL)



RL holds great promise in the next era of artificial intelligence.

One more: RL for thinking models

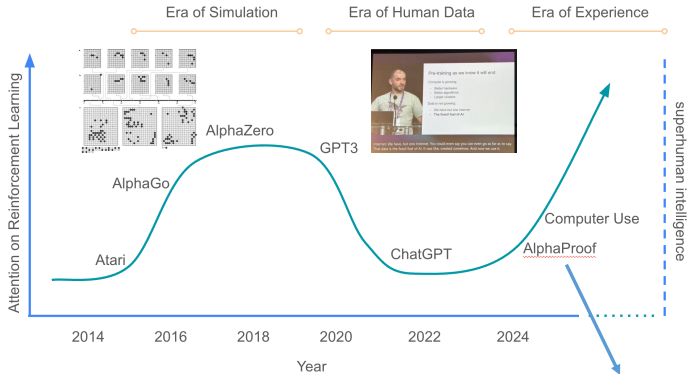


Turing Award Goes to 2 Pioneers of Artificial Intelligence

Andrew Barto and Richard Sutton developed reinforcement learning, a technique vital to chatbots like ChatGPT.



The cycles of RL



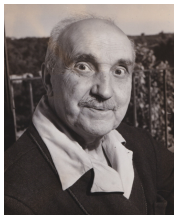
by human mathematicians, AlphaProof's reinforcement learning (RL) algorithm¹ subsequently generated a hundred million more through continual interaction with a formal proving system. This focus on interactive experience allowed AlphaProof to explore mathematical possibilities beyond the confines of pre-existing formal proofs, so as to discover solutions to novel and challenging problems. Informal mathematics has also achieved success by replacing expert generated data with self-generated data; for example, recent work

Figure credit: [Silver and Sutton, 2025]

Sequential decision making

*“Those who cannot remember the past are
condemned to repeat it.”*

—George Santayana

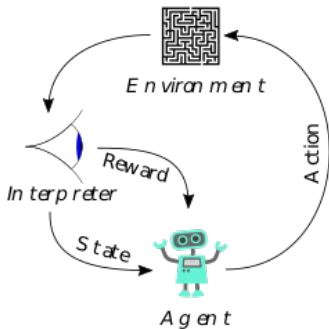


- Games
- Robotics navigation and control
- Pricing and supply chain management
- Recommendation systems
- Portfolio optimization
- Large language models and agents

Learn from past to predict and optimize future performance

Reinforcement learning

In reinforcement learning (RL), an agent learns through interaction with the (unknown) environment.



RL has deep connection with control theory, and is also sometimes called **approximate dynamic programming**. It can be viewed as a type of **optimal control theory** with no pre-defined model of the environment.

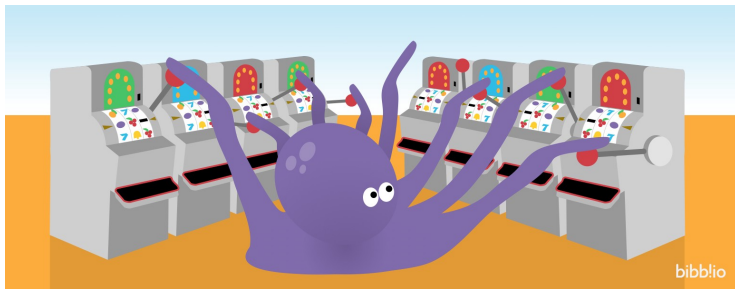
Challenges of RL

- explore or exploit: unknown or changing environments
- credit assignment problem: delayed rewards or feedback
- enormous state and action space
- nonconvex optimization



Multi-arm bandit

Which slot machine will give me the most money?



Learning the best arm

Can we **learn** which slot machine gives the most money?



\$1
\$0
\$0



\$1
\$4
\$0
\$2
\$1
\$3
\$5



\$1
\$0
\$1
\$2

Learning the best arm via trial-and-error

Which arm do I pick next, so that I maximize my reward over time?



\$1

\$0

\$0



\$1

\$4

\$0

\$2

\$1

\$3

\$5



\$1

\$0

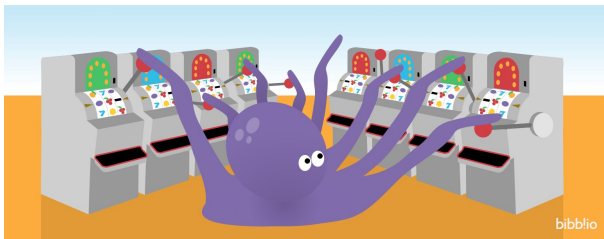
\$1

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Exploration-exploitation trade-off



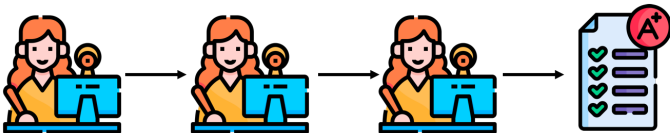
Which arm should I play?

- Best arm observed so far? (exploitation)
- Or should I look around to try and find a better arm? (exploration)

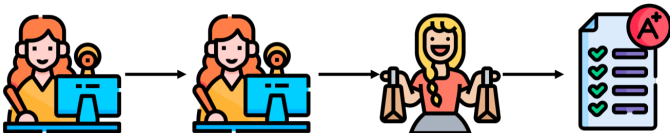
We need both in order to maximize the total reward.

Credit assignment problem

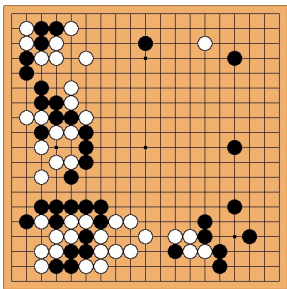
What is the action that leads to the desired outcome?



What if....

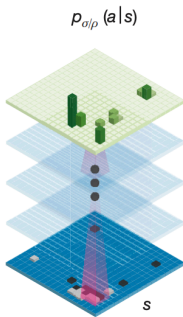


Enormous problem size and function approximation



$$S \approx 2 \cdot 10^{170}$$

Policy network



Value network

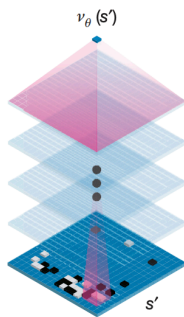
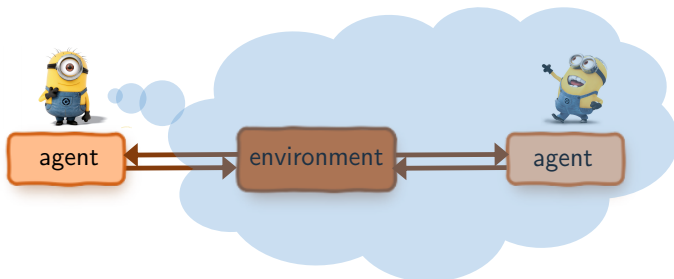


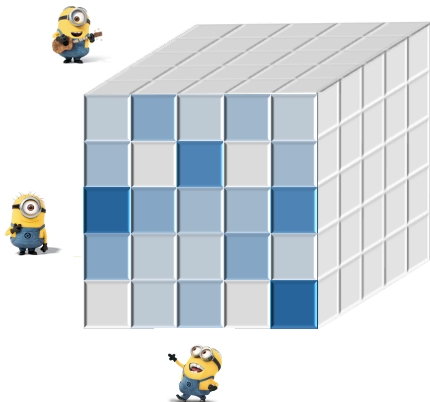
Figure credit: AlphaGo

Challenges in MARL: nonstationarity



From a single-agent perspective:
the environment is **time-varying** and **nonstationary**!

Challenges in MARL: curse of multiple agents



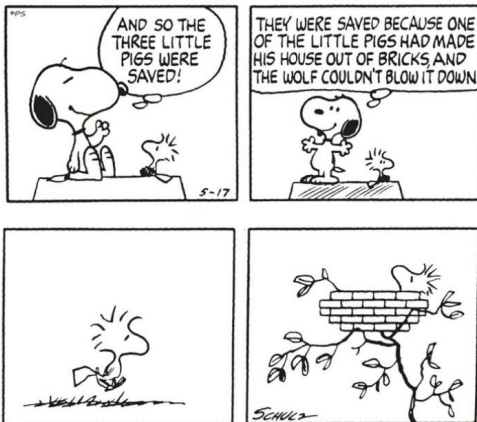
The explosion of choices:
The joint action space grows **exponentially** with the agents!

Partial observability in RL



Goal of this course

- Build the “foundations” but with an eye on modern applications
- Research-oriented
- models, algorithms and their analyses



Sample efficiency

Collecting data samples might be expensive or time-consuming



clinical trials



autonomous driving

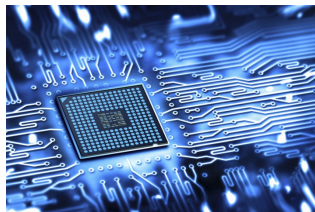
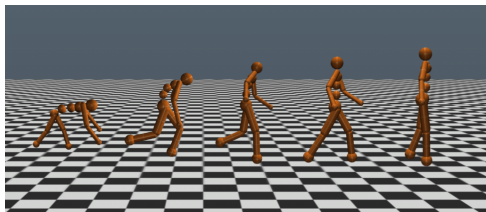


online ads

Calls for design of sample-efficient RL algorithms!

Computational efficiency

Running RL algorithms might take a long time and space



many CPUs / GPUs / TPUs + computing hours

Calls for computationally efficient RL algorithms!

From asymptotic to non-asymptotic analyses



Non-asymptotic analyses are key to understand sample and computational efficiency in modern RL.

Logistics

Basic information

- Class: Tuesdays, 9:25 – 11:20 am
- Class location: Watson Center A74
- Instructor: Yuejie Chi, yuejie.chi@yale.edu
- Instructor's office hours: Tuesdays, 11:30 am–12:30 pm (right after class)
- Announcements, readings, and logistics: canvas, Ed, and <https://yuejiechi.github.io/sds6190.html>
- Project submissions: Gradescope, unless otherwise noted

Course format and expectations

- Class meets once a week
- Class meetings combine
 - lectures
 - discussion
 - student presentations
- A substantial final project running throughout the semester (more on this later)
- Research-oriented exploration of theoretical foundations and modern applications

Tentative topics

- Stochastic and adversarial bandits
- Markov decision processes and dynamic programming
- Evaluation and planning with a simulator
- Online reinforcement learning
- Imitation learning and offline reinforcement learning
- Policy optimization
- Multi-agent reinforcement learning
- RL for reasoning and alignment of foundation models: RLHF and RLVR
- Cross-cutting themes: sample complexity, computational efficiency, and function approximation

Useful references

We recommend these books and notes, but will not follow them closely ...

- **Reinforcement Learning: Theory and Algorithms**, by Alekh Agarwal, Nan Jiang, Sham M. Kakade, and Wen Sun
- **Reinforcement Learning: An Introduction**, by Richard S. Sutton and Andrew G. Barto
- **Reinforcement Learning and Optimal Control**, by Dimitri P. Bertsekas
- **Bandit Algorithms**, by Tor Lattimore and Csaba Szepesvari
- **Sample Complexity of Reinforcement Learning: A Non-Asymptotic Perspective**, by Yuxin Chen, Yuejie Chi, Jianqing Fan, Gen Li, Yuting Wei, and Yuling Yan

More references will be provided at each lecture.

Prerequisites

- Basic fluency in
 - linear algebra and probability
 - optimization and machine learning
 - programming
- Prior exposure to reinforcement learning is helpful, but not required
- *Concentration inequalities* are a plus, but not necessary
- If you are unsure about your preparation, please ask early

Grading

- Participation and attendance: 5%
- In-class quizzes: 15%
 - ~10 in total, on canvas at the end of each lecture, you can drop 2.
- Midterm presentation: 10%
- Project development: 35%
- Final deliverables and report: 35%

The project

- Pursue an **original research direction** related to modern reinforcement learning and sequential decision making
- Develop the project throughout the semester through
 - proposal drafts and a final proposal
 - research checkpoints
 - peer review and feedback
- Present the project in class
- Submit final deliverables and a report
- Project development (35%) and final deliverables (35%) together account for **70%** of the course grade

Key project and presentation milestones

September–October

- Sep. 22: project proposal, first draft
- Oct. 6: project proposal, second draft
- Oct. 13: midterm presentation draft
- Oct. 20: midterm presentation and peer feedback
- Oct. 27: project proposal, final version

November–December

- Nov. 10: project milestone 1
- Nov. 24: project milestone 2
- Dec. 1: final presentation draft
- Dec. 8: final project presentation
- Dec. 15–22: final project report due

The schedule is provisional and may be updated as the semester proceeds.

Rationale of the project schedule

- You should have “actionable target” in mind while taking the course.
 - You are encouraged to connect the project to something you are already working on in your research.
- Start early, but allow iterations to refine ideas.
- Read topics before they are “covered”, and come to lectures with questions.
- Require strong maturity to succeed.

The use of LLMs

You are encouraged to use LLMs to help with

- writing and understanding the lecture materials
- brainstorming
- coding

You own your submitted or posted work and take full responsibility for it.

- Verify the output critically
- Be ready to defend your work when prompted
- Disclose how LLMs were used in submitted work

Enrollment

Enrollment requires faculty approval.

- Students interested in this course should submit a statement of interest, describing their background in relevant courses.
- This should be submitted on Canvas in the “Quiz” area.
- You will be informed within a week after the first meeting of the class.

For the first offering, the target class size is about 20.

Let's get started!

Enjoy Yourself

Stochastic bandits

A/B testing

How do you decide which variation leads to higher traffic/revenue?



Figure credit: internet.

A/B testing: explore each variation equally first, then deploy the statistically better one.

From A/B testing to multi-arm bandits

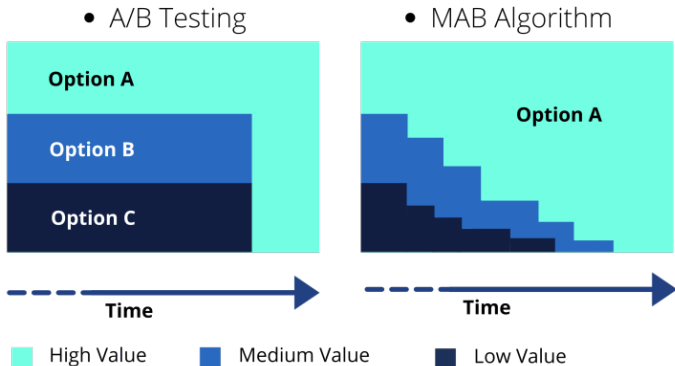
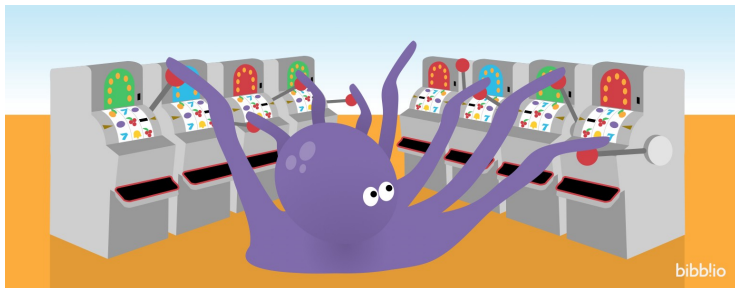


Figure credit: internet.

Multi-arm bandits: simultaneous exploration and exploitation, dynamic allocation.

Multi-arm bandit

Which slot machine will give me the most money?



First proposed in [Thompson, 1933], popularized by [Robbins, 1952]

Learning the best arm

Can we **learn** which slot machine gives the most money?



\$1
\$0
\$0



\$1
\$4
\$0
\$2
\$1
\$3
\$5



\$1
\$0
\$1
\$2

Formulation

We can play multiple rounds $t = 1, 2, \dots, T$.

In each round, we **select an arm** i_t from a fixed set $i = 1, 2, \dots, n$; and **observe the reward** r_t that the arm gives.

Arm 1



Arm 2

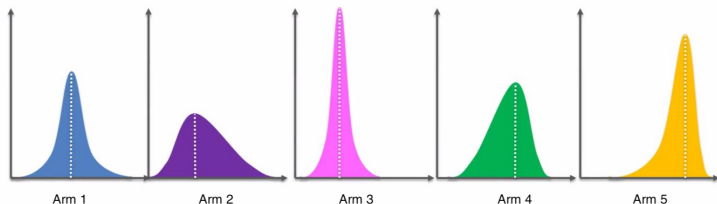


Arm 3



Objective: Maximize the total reward over time.

Stochastic bandit



- The reward at each arm is **stochastic** (e.g., 1 with probability p_i and otherwise 0).
- Suppose the rewards are independent over time. The **best arm** is then the arm with highest expected reward.

Example of online ads: arm = ad, reward = 1 if the user clicks on the ad and 0 otherwise

Stochastic bandit with i.i.d. rewards

We consider a simple setting with i.i.d. bounded rewards.

- Each arm distributes rewards according to some (unknown) distribution over $[0, 1]$, with

$$\mathbb{E}[r_{i,t}] = \mu_i, \quad \forall i \in [n], t = 1, 2 \dots$$

- Suppose we play arm i_t at round t , and receive the reward

$$r_{i_t,t}$$

drawn i.i.d. from the arm i_t 's distribution.

Partial information: Every round we cannot observe the reward of all arms: we just know the reward of the arm that we played.

Regret: performance metric

We design algorithms that determine the sequence $\{i_t\}$, i.e. *policies*.

How to evaluate the performance?

Definition 1 (Expected regret)

The **expected regret over T rounds** is defined as

$$R_T = \max_{1 \leq i \leq n} \mathbb{E} \left[\sum_{t=1}^T (r_{i,t} - r_{i_t,t}) \right] = T\mu^* - \mathbb{E} \left[\sum_{t=1}^T r_{i_t,t} \right],$$

where $\mu^* = \max_{1 \leq i \leq n} \mu_i$ is the highest expected reward over all arms.

- 1st term captures the highest cumulative reward in *hindsight*.
- 2nd term captures the *actual* accumulated reward.

Regret decomposition lemma

Since $\mathbb{E}[r_{i_t,t}] = \mathbb{E}[\sum_{i=1}^n \mu_i \mathbb{I}_{i_t=i}] = \sum_{i=1}^n \mu_i (\mathbb{E} \mathbb{I}_{i_t=i})$, then

$$\begin{aligned} R_T &= \sum_{t=1}^T \left[\sum_{i=1}^n \mu^* (\mathbb{E} \mathbb{I}_{i_t=i}) - \sum_{i=1}^n \mu_i (\mathbb{E} \mathbb{I}_{i_t=i}) \right] \\ &= \sum_{i=1}^n \Delta_i \mathbb{E} \left[\sum_{t=1}^T \mathbb{I}_{i_t=i} \right] \\ &=: \sum_{i=1}^n \Delta_i \mathbb{E} [T_{i,T}] \end{aligned}$$

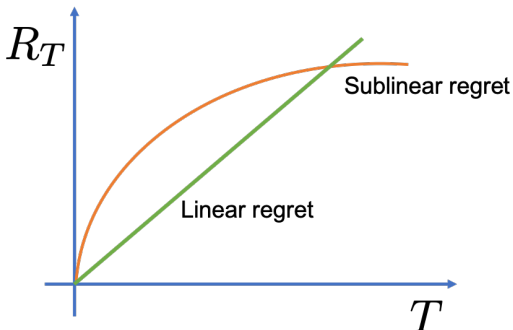
where

- $\Delta_i = \mu^* - \mu_i$ is the sub-optimality gap of arm i ;
- $T_{i,T} = \sum_{t=1}^T \mathbb{I}_{i_t=i}$ is the number of times arm i is played in T rounds.

Sublinear regret

Sublinear regret: most MAB algorithms aim to achieve **sublinear regret**, so that the average regret goes to 0 as $T \rightarrow \infty$:

$$\lim_{T \rightarrow \infty} \frac{R_T}{T} = 0$$



From ϵ -greedy to UCB algorithm

Learning the best arm via trial-and-error

Which arm do I pick next, so that I maximize my reward over time?



\$1

\$0

\$0



\$1

\$4

\$0

\$2

\$1

\$3

\$5



\$1

\$0

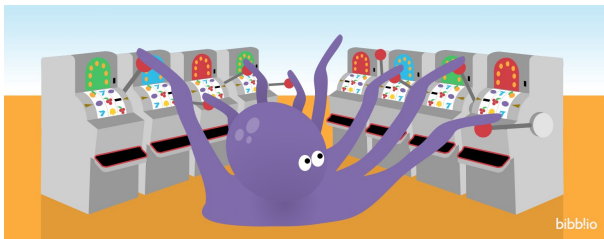
\$1

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Exploration-exploitation trade-off



Which arm should I play?

- Best arm observed so far? (exploitation)
- Or should I look around to try and find a better arm? (exploration)

We need both in order to maximize the total reward.

An ϵ -greedy approach

Exploit, but **explore a random arm ϵ fraction of the time.**

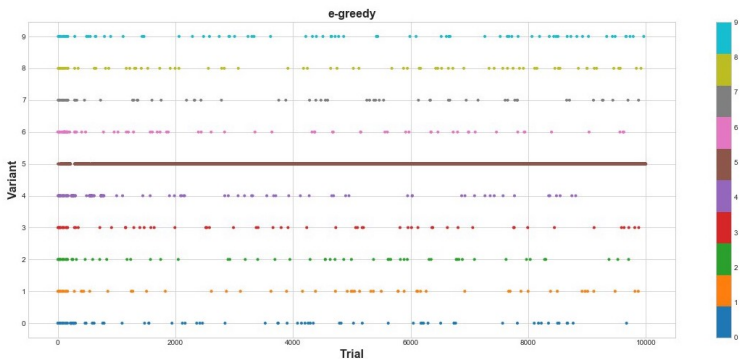
- ① **Initial phase:** Try each arm and observe the reward.
- ② For each round $t = n + 1, \dots, T$:
 - Calculate the empirical average reward for each arm i :

$$\bar{\mu}_{i,t} = \frac{\text{total reward from pulling this arm in the past}}{\text{number of times I pulled this arm}} = \frac{\sum_{t:i_t=i} r_t}{\sum_{t:i_t=i} 1},$$

where i_t is the index of the arm played at time t , r_t is the reward.

- With probability $1 - \epsilon$, play the arm with highest $\bar{\mu}_{i,t}$ and observe the reward. Otherwise, choose an arm at random and observe the reward.

Understanding ϵ -greedy



- In the first thousand iterations, all arms are chosen fairly frequently.
- Eventually the algorithm realizes that arm 5 has the highest expected reward.

Regrets of greedy policies

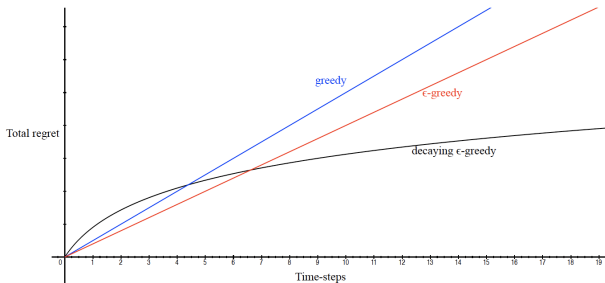


Figure credit: David Silver's lecture.

- Greedy policy incurs linear regret since it can lock on a sub-optimal policy.
- ϵ -greedy always explores by ϵ fraction and therefore its regret is still linear (recall the regret decomposition lemma).
- Decaying ϵ helps, however it is hard to design the schedule.

The UCB algorithm

[Auer et al., 2002]: the idea is to **always try the best arm**, where “best” includes exploration and exploitation.

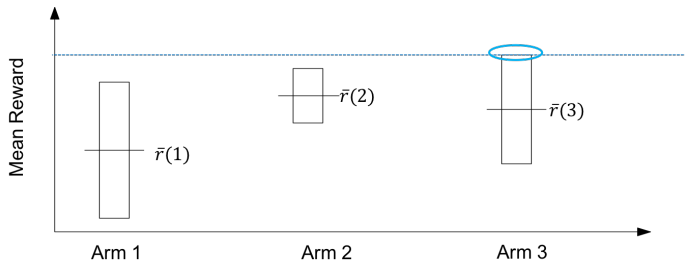
- ➊ **Initial phase:** try each arm and observe the reward.
- ➋ For each round $t = n + 1, \dots, T$:
 - Calculate the **UCB (upper confidence bound) index** for each arm i :

$$\text{UCB}_{i,t} = \bar{\mu}_{i,t} + \sqrt{\frac{\log t}{T_{i,t}}},$$

where $\bar{\mu}_{i,t}$ is the empirical average reward for arm i and $T_{i,t}$ is the number of times arm i has been played up to round t .

- Play the arm with the highest UCB index and observe the reward.

Understanding UCB



$$UCB_{i,t} = \bar{\mu}_{i,t} + \sqrt{\frac{\log t}{T_{i,t}}},$$

- **Exploitation:** $\bar{\mu}_{i,t}$ is the average observed reward. High observed rewards of an arm leads to high UCB index.
- **Exploration:** $\sqrt{\frac{\log t}{T_{i,t}}}$ decreases as we make more observations ($T_{i,t}$ grows). Few observations of an arm leads to high UCB index.

Theory of UCB algorithm

Theorem 2 (Instance-dependent regret bound of UCB)

For $T \geq n$, the expected regret of UCB algorithm is upper bounded as

$$R_T \leq \sum_{i: \Delta_i > 0} \left(\frac{4 \log T}{\Delta_i} + 8 \Delta_i \right) \leq \sum_{i: \Delta_i > 0} \frac{4 \log T}{\Delta_i} + 8n,$$

where $\Delta_i = \mu^* - \mu_i$ is the sub-optimality gap of arm i .

- The regret bound scales with the *harmonic mean* of the gaps,

$$R_T \lesssim \frac{n \log T}{\text{harmonic mean}(\{\Delta_i\})}.$$

- E.g. $\Delta_2 = \frac{1}{2}$, $\Delta_3 = \frac{1}{2}$, harmonic mean = $\frac{1}{2}$.
- E.g. $\Delta_2 = \frac{1}{10}$, $\Delta_3 = \frac{1}{2}$, harmonic mean = $\frac{1}{6}$.
- When Δ_i 's are constants, the regret scales as (ignoring n)

$$R_T = O(\log T),$$

which is nearly the best we can hope for! (We'll see why later.)

Gap-free bound of UCB algorithm

The gap-dependent bound may become too loose when Δ_i is, say, asymptotically small, $\Delta_i \sim \log T/T$.

Fortunately, this can be fixed by studying the following **instance-independent (aka worst-case) bound**.

Theorem 3 (Instance-independent regret bound of UCB)

For $T \geq n$, the expected regret of UCB algorithm is upper bounded as

$$R_T \leq 4\sqrt{nT \log T} + 8n.$$

- When $n = O(1)$, the regret scales as

$$R_T = O(\sqrt{T \log T}) = \tilde{O}(\sqrt{T})$$

- The logarithmic factor can be shaved away [Audibert and Bubeck, 2009].

Analysis

Toolkit: Hoeffding's inequality

Theorem 4 (Hoeffding's inequality)

Let $X_1, X_2, \dots, X_n$ be independent random variables satisfying $a_i \leq X_i \leq b_i$. Then for all $\delta \geq 0$,

$$\mathbb{P} \left(\left| \sum_{i=1}^n X_i - \mathbb{E} \left[\sum_{i=1}^n X_i \right] \right| \geq \varepsilon \right) \leq 2 \exp \left(- \frac{2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

Setting $a_i = 0$, $b_i = 1$, and $\mathbb{E}[X_i] = \mu$, we obtain

$$\begin{aligned} \mathbb{P} \left(\left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| \geq \varepsilon \right) &\leq 2 \exp(-2n\varepsilon^2) \\ \Rightarrow \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right| &\leq \sqrt{\frac{\log(2/\delta)}{2n}} \quad \text{with prob. } 1 - \delta. \end{aligned}$$

This will allow us to talk about how the mean reward concentrates around the true mean.

Implications of Hoeffding's inequality

For each arm i at time t , with probability at least $1 - 2/t^2$,

$$|\bar{\mu}_{i,t} - \mu_i| < \sqrt{\frac{\log t}{T_{i,t}}}$$
$$\Rightarrow \text{UCB}_{i,t} = \bar{\mu}_{i,t} + \sqrt{\frac{\log t}{T_{i,t}}} \geq \mu_i.$$

Optimism in the face of uncertainty:

acting according to the UCB index, which is an upper bound of the true mean μ_i .



Bound the number of sub-optimal pulls

Recall that

$$R_T = \sum_{i=1}^n \Delta_i \underbrace{\mathbb{E}[T_{i,T}]}_{\text{control target}}.$$

Key observation: at each t , the UCB index of the sub-optimal arms $i \neq i^*$ will be sufficiently apart from the optimal one and arm i will not get pulled (i.e. $i_{t+1} \neq i$), as long as $T_{i,t}$ is sufficiently large:

$$\begin{aligned} \text{UCB}_{i,t} = \bar{\mu}_{i,t} + \sqrt{\frac{\log t}{T_{i,t}}} &\leq \mu_i + 2\sqrt{\frac{\log t}{T_{i,t}}} && \text{(Hoeffding)} \\ &\leq \mu_{i^*} \leq \text{UCB}_{i^*,t} && \text{(optimism/Hoeffding)} \end{aligned}$$

as long as

$$T_{i,t} \geq \frac{4 \log t}{\Delta_i^2}$$

with probability at least $1 - 4/t^2$ (we applied Hoeffding twice).

Bound the number of sub-optimal pulls

$$\begin{aligned}\mathbb{E}[T_{i,T}] &\leq \mathbb{E}\left[\sum_{t=0}^{T-1} \mathbb{I}(i_{t+1} = i)\right] \\&= \mathbb{E}\left[\sum_{t=0}^{T-1} \mathbb{I}\left(i_{t+1} = i, T_{i,t} < \frac{4 \log t}{\Delta_i^2}\right)\right] + \mathbb{E}\left[\sum_{t=0}^{T-1} \mathbb{I}\left(i_{t+1} = i, T_{i,t} \geq \frac{4 \log t}{\Delta_i^2}\right)\right] \\&\leq \frac{4 \log T}{\Delta_i^2} + \sum_{t=n}^{T-1} \mathbb{P}\left(i_{t+1} = i, T_{i,t} \geq \frac{4 \log t}{\Delta_i^2}\right) \\&\leq \frac{4 \log T}{\Delta_i^2} + \sum_{t=n}^{T-1} \mathbb{P}\left(i_{t+1} = i \mid T_{i,t} \geq \frac{4 \log t}{\Delta_i^2}\right) \mathbb{P}\left(T_{i,t} \geq \frac{4 \log t}{\Delta_i^2}\right) \\&\leq \frac{4 \log T}{\Delta_i^2} + \sum_{t=n}^{T-1} \frac{4}{t^2} \\&\leq \frac{4 \log T}{\Delta_i^2} + 8.\end{aligned}$$

A key lemma

Lemma 5 (bounding the number of pulls of sub-optimal arms)

For any arm with $\Delta_i > 0$, it holds that

$$\mathbb{E}[T_{i,T}] \leq \frac{4 \log T}{\Delta_i^2} + 8.$$

Proof of Theorem 2:

$$\begin{aligned} R_T &= \sum_{i=1}^n \Delta_i \mathbb{E}[T_{i,T}] \leq \sum_{\Delta_i > 0} \Delta_i \left(\frac{4 \log T}{\Delta_i^2} + 8 \right) \\ &= \sum_{\Delta_i > 0} \left(\frac{4 \log T}{\Delta_i} + 8 \Delta_i \right). \end{aligned}$$

From gap-dependent to gap-independent bounds

Intuition: for some Δ to be determined later,

- For arms $\{i : \Delta_i \geq \Delta\}$ with **large gaps**: use the gap-dependent bound

$$\mathbb{E}[T_{i,T}] \leq \frac{4 \log T}{\Delta_i^2} + 8;$$

- For arms $\{i : \Delta_i < \Delta\}$ with **small gaps**: use the naive bound

$$\sum_i \mathbb{E}[T_{i,T}] \leq T.$$

Hence,

$$\begin{aligned} R_T &= \sum_{i=1}^n \Delta_i \mathbb{E}[T_{i,T}] \leq \sum_{i: \Delta_i \geq \Delta} \Delta_i \left(\frac{4 \log T}{\Delta_i^2} + 8 \right) + \sum_{i: \Delta_i < \Delta} \Delta_i \mathbb{E}[T_{i,T}] \\ &\leq \frac{4n \log T}{\Delta} + 8n + \Delta T. \end{aligned}$$

Choosing $\Delta = \sqrt{\frac{4n \log T}{T}}$, we obtain $R_T \leq 4\sqrt{nT \log T} + 8n$.

Wrap-up

Quiz 1: on canvas and submit why you are eager to take this class!

Due date: this Friday.

References I



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